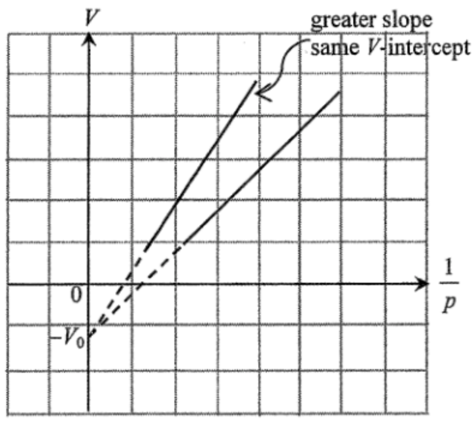
Q1

1.	(a)	- Put the sphere into the water bath for a few minutes - Put / transfer the sphere into the polystyrene cup (of water) - Measure the final / maximum temperature T_f of the water with a thermometer	1A 1A 1A	
		$0.80 \times c_b \times (80 - T_f) = 0.50 \times 4200 \times (T_f - T_0)$	1A	
		$c_b = 2625 \times \dfrac{T_f - T_0}{80 - T_f}$ (J kg^{-1} °C^{-1})		
		Precautions: - Dry the sphere with the towel quickly before putting it into the cup - Make sure the sphere is fully immersed in water - Stir the water thoroughly Any **TWO**	 1A 1A 1A	
				6
	(b)	Thermal energy / heat is lost during the transfer / drying of the sphere	1A	
		Or Thermal energy / heat absorbed by thermometer, stirrer or cup	1A	
		Or The temperature of the sphere is higher than T_f when this final temperature is measured (i.e. T_f not reaching its maximum)	1A	
		Thus temperature rise of water in the cup is lower than it should be.	1A	
				2

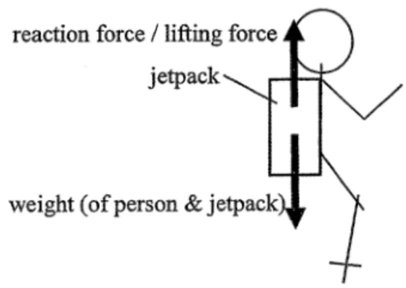
Q2

2. (a) $pV = nRT$	1M	
$(1.0\times10^{5})(6.0\times10^{-5}) = n\ (8.31)(273+25)$	1M	
$n = 2.422891 \times 10^{-3}$ moles $\approx 2.42 \times 10^{-3}$ moles		
No. of molecules $= n\,N_A$ $= n \times 6.02 \times 10^{23}$ $= 1.458581 \times 10^{21} \approx 1.46 \times 10^{21}$	1A	
Alternative method:		
$pV = nRT = \left(\dfrac{N}{N_A}\right)RT \Rightarrow N = \left(\dfrac{pV}{RT}\right)N_A$	1M	
$N = \dfrac{(1.0\times10^{5})(6.0\times10^{-5})}{(8.31)(273+25)} \times (6.02\times10^{23})$	1M	
$= 1.458581 \times 10^{21} \approx 1.46 \times 10^{21}$	1A	
	3	
(b) (i) - The piston should be pushed or pulled slowly / lightly. (Any **ONE**)	1A	
- Avoid taking readings immediately after moving the piston.	1A	
- The syringe should not be grasped by hand too long (when the piston is pushed in or pulled out).	1A	
	1	
(ii) V_0 - volume of gas trapped in the rubber tubing / space connecting the pressure sensor and syringe.	1A	
	1	
(iii) 	1A 1A	**NO** mark for a curve
	2	

Q3

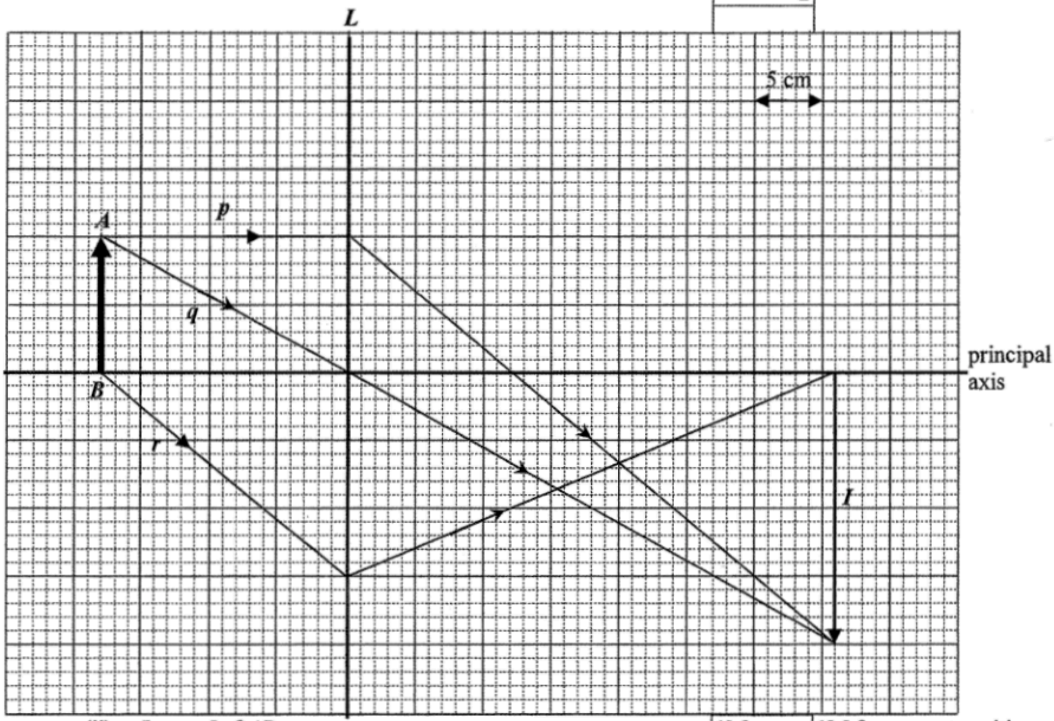
3. (a) $a = \dfrac{6-0}{2-0}$	1M	
$= 3\ \text{m s}^{-2}$ (downwards)	1A	
	2	
(b) A: 395 N B: 569 N C: 685 N	1A	
In stage B, balance reading = weight (Newton's 1st law),		
$mg = m \times 9.81\ \text{m s}^{-2} = 569\ \text{N}$	1M	
$m = 58.0\ \text{kg}$	1A	For $g = 10\ \text{m s}^{-2}$, $m = 56.9\ \text{kg}$
<u>Or</u> In stage A, according to Newton's 2nd law		
$(569 - 395)\ \text{N} = ma = m(3\ \text{m s}^{-2})$	1M	
$m = 58.0\ \text{kg}$	1A	
	3	
(c) (i) For stage C, by Newton's 2nd law,		
$F = ma$		
$(569 - 685)\ \text{N} = (58.0\ \text{kg})\ a$	1M	
$a = -2\ \text{m s}^{-2}$		For $g = 10\ \text{m s}^{-2}$, $m = 56.9\ \text{kg}$, $a = -2.04\ \text{m s}^{-2}$
Thus, $a = \dfrac{0-6}{T-12} = -2\ \text{m s}^{-2}$	1M	
$T = 15\ \text{s}$		
	2	
(ii) Height $\approx$ displacement of lift = area under graph	1M	
$= \dfrac{(12-2)+15}{2} \times 6$		
$= 75\ \text{m}$	1A	
	2	

Q4

4. (a) According to Newton's second law of motion, a (net) force acts on the water so as to change its momentum. (Or magnitude of the force equals the rate of change of momentum of water).	1A	
According to Newton's third law of motion, a force acting downwards on the ejecting water (by the jetpack), the water exerts a reaction (equal but upward / opposite) on the jetpack / person as well.	1A 1A	
	3	
(b) 	1A	
	1	
(c) (i) $F = \frac{\Delta p}{\Delta t} = \frac{\Delta m}{\Delta t} \times (\vec{v} - \vec{u})$	1M	
$\frac{\Delta m}{\Delta t} \times (10 - (-10))\ \text{m s}^{-1} = 1000\ \text{N}$ $\frac{\Delta m}{\Delta t} = 50\ (\text{kg s}^{-1})$	1A	Accept using kg as the unit.
	2	
(ii) $(\frac{\Delta m}{\Delta t})gh + \frac{1}{2}(\frac{\Delta m}{\Delta t}) \times v^2$	1M	Accept using m for $(\frac{\Delta m}{\Delta t})$
$= (50\ \text{kg s}^{-1})(9.81\ \text{m s}^{-2})(7.5\ \text{m}) + \frac{1}{2}(50\ \text{kg s}^{-1}) \times (10\ \text{m s}^{-1})^2$	1M	
$= 6178.75$ W or 6.17875 kW	1A	For $g = 10\ \text{m s}^{-2}$, 3750 W + 2500 W = 6250 W
	3	
(d) The same as the same lifting force / (water) jet speed is needed.	1A 1A	
	2	

Q5

5.	(a)	(i)	Convex (lens)	1A		
			Only convex lens forms real image (which can be captured by a screen)	1A		
			Or Concave lens always forms virtual image (which cannot be captured by a screen).	1A		
			Or The image is formed on the other side of the lens.	1A		
					2	
		(ii)	 opaque screen	1A		
					1	
	(b)	(i)	Image distance $v = 54 - 18 = 36$ cm ($D = 54$ cm)	1A		
			Magnification $= \frac{v}{u} = \frac{36}{18} = 2$	1M/1A		
					2	



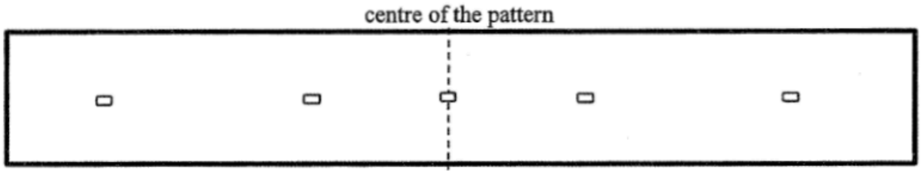
		(ii)	Image I of AB	1M		1M for correct position of I
			Rays p and q	1M		
			Ray r	1M		
					3	
		(iii)	Focal length = 12 ± 0.5 cm	1M/1A		
					1	
		(iv)	Move the lens 18 cm farther away from the object.	1M		
			Or move the lens 18 cm closer to / towards the screen.	1M		
			Height ratio = 1 : 4.	1A		
					2	

Q6

6. (a) (i)	$$\Delta y = \frac{\lambda D}{a}$$ $$\frac{(4.0-0)\times 10^{-2}}{10} = \frac{\lambda(1.8)}{0.3\times 10^{-3}}$$	1M+1M		
	$\lambda = 6.666667 \times 10^{-7}$ m $\approx 6.67 \times 10^{-7}$ m or 667 nm	1A		
			3	
(ii)	To ensure that the light through the 2 slits diffracts enough to interfere / overlap.	1A 1A		Explanation in terms of formula $\lambda = \Delta y \frac{a}{D}$ is **NOT** accepted as a is the slit separation given, which is fixed.
			2	
(b) (i)	$d \sin\theta = m\lambda$ $\frac{10^{-3}}{500}\sin\theta = 6.67 \times 10^{-7}$ m	1M		
	$\theta = 19.471221° \approx 19.47°$			
	Separation between central and first-order bright spot $= 1.8 \tan 19.47°$	1M		
	$= 0.636396$ m ≈ 0.636 m	1A		
			3	

(ii)

centre of the pattern



symmetry about the central bright spot (2nd order shown)	1A	
separation between 2nd and 1st -order bright spots is larger	1A	
		2

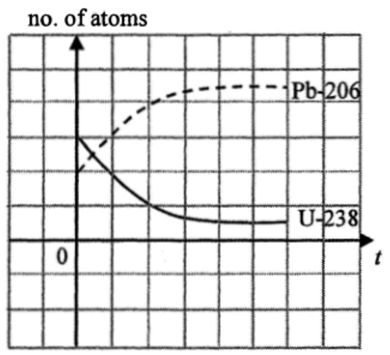
Q7

7. (a) (i)	$R = 10\text{ k}\Omega$ (circuit I) $V = \dfrac{(\frac{1}{10\text{ k}\Omega}+\frac{1}{10\text{ k}\Omega})^{-1}}{10\text{ k}\Omega+(\frac{1}{10\text{ k}\Omega}+\frac{1}{10\text{ k}\Omega})^{-1}}\times 6\text{ V}$ $= 2\text{ V}$ $R = 100\ \Omega$ (circuit II) $V = \dfrac{(\frac{1}{100\Omega}+\frac{1}{10\text{k}\Omega})^{-1}}{100\Omega+(\frac{1}{100\Omega}+\frac{1}{10\text{k}\Omega})^{-1}}\times 6\text{ V}$ $= 2.985\text{V}$	1M 1A 1A	3	1M for appropriate method in calculating voltage Note: 100 Ω and 10 kΩ in parallel ≈ 99.0099 Ω Accept stating that V slightly < 3 V
(ii)	Resistance of circuit / that part of circuit would be lowered / altered significantly when introducing the voltmeter (i.e. loading effect). <u>Or</u> The resistance of the voltmeter is comparable to the resistance of resistor R. Resistance of voltmeter should be much higher than the resistance of the part of the circuit under study.	1A 1A 1A	2	
(b) (i)	V_m does **NOT** give the true voltage for the resistor. $R_m = R_A + R$	1A 1A	2	
(ii)	For circuit III $R_m = R + R_A = 10 + 1 = 11\ \Omega$ percentage error $= \dfrac{1\Omega}{10\Omega} \times 100\ \%$ $= 10\%$	1M 1A	2	

Q8

8.	(a)	(i)	- the air loses its insulating properties	1A		
			Or electrons or ions can pass through the air (between clouds and Earth or between clouds and clouds)	1A		
					1	
		(ii)	$E = \frac{V}{d}$	1M		
			$V = E\,d = (3 \times 10^5) \times 2000$			
			$= 6 \times 10^8$ V	1A		
					2	
	(b)	(i)	magnetic field into paper (due to upward lightning current)	1A		
			$B = \frac{\mu_0 I}{2\pi r}$	1M		
			$= \frac{4\pi \times 10^{-7} \times 30000}{2\pi \times 1500}$			
			$= 4 \times 10^{-6}$ T	1A		
					3	
		(ii)	When the lightning current is increasing, the induced current flows in the anticlockwise direction so as	1A		
			to oppose the increasing magnetic field (into paper).	1A		
			After reaching maximum, the lightning current is decreasing, the induced current flows in the clockwise / opposite direction.	1A		
					3	
		(iii)	Electric field (in the atmosphere).	1A		
			E-field increases / builds up (to threshold) before lightning happens	1A		
			Or Lightning current and magnetic field only exist during lightning.	1A		
					2	

Q9

9. (a) $(4)n_\alpha = 238 - 206 \Rightarrow n_\alpha = 8$	1A	
$(2)n_\alpha + (-1)n_\beta = 92 - 82 \Rightarrow n_\beta = 6$	1A	
	2	
(b) (i) $N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$	1M	
$\frac{3}{5} = \left(\frac{1}{2}\right)^{t/4.5\times10^9\,\text{yr}}$		
<u>Or</u> $N = N_0\, e^{-\lambda t}$ and $\lambda = \frac{\ln 2}{T_{1/2}}$	1M	
$\therefore t = 3.316 \times 10^9$ years $\approx 3.3 \times 10^9$ years	1A	
	2	
(ii) Answer in (i) is an underestimate. The original number of U-238 atoms should be greater.	1A	
$\therefore$ the ratio $\frac{\text{present number of U-238 atoms}}{\text{original number of U-238 atoms}} = \frac{N_t}{N_0}$ is in fact smaller (than $\frac{3}{5}$), thus longer time should have been elapsed.	1A	Accept 'more U decayed'.
	2	
(iii) 	2A	
	2	